(i) Effect of reducing movement.

Hypothesis: The peak gets lower and the arrival is later.

Rationale: The change in the movement speed will effectively change how often an agent meets someone new. With less movement, infected agents stay near agents they have already exposed. Hence, the disease spreads more like a slow front, rather than random mixing.

(ii) Consistency across simulated runs

Hypothesis: Each run would be different, and is up to chance.

Rationale: With only 5 infected agents, the spread could run off from the start with by-chance encounters. It could also be that the infected agents die before they can expose others. After running many simulations, the distribution of

the peak number and the time elapsed til the peak should be unimodal.

(iii) Average agent-based curves $\approx$ SIR curve

Hypothesis: The curve would look similar, but the epidemic will be slower and smaller.

Rationale: Main difference is that SIR assumes full mixing. Second, on the grid, the contacts are local and correlated (the infected agents would end up clustering) Hence, the spread will only occur at the edges of the cluster, making the growth close to t^2 than exponential. As a note, this is different from real life scenario, where people travel long distances during trips and conferences, which would make the spread closer to exponential.